



RELEASE HISTORY

Power Glove Vision Changelog

Versioned features, fixes, security changes, and documentation updates.

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This file records user-visible PowerGlove Vision changes. The project follows [Keep a Changelog](#) categories and uses [Semantic Versioning](#). Git remains the authoritative record for line-level and file-level history.

[Unreleased]

Added

- Version-2 HMAC controller messages with receiver-issued challenges, restart/retired-session replay rejection, and bounded nonblocking handshakes. Existing native and gamepad mappings are unchanged; upgrade both computers together.
- Per-player saved calibration with explicit same-position reuse after switching; internal stores migrate to version 4 with private recovery backups.
- Version-2 hand backups now include all effective sensitivity pairs and software identity. Restore separately reviews complete sensitivity and calibration reuse. Earlier version-2 files work; version-1 portable exports are no longer accepted.
- Background hostname refresh removes synchronous DNS/mDNS work from controller sends without queuing frames or states. An isolated 12-lookup Controller sample measured 2.35 ms median and 107.54 ms maximum cold-cache lookup time before this change.
- Independent host Wi-Fi health in Setup and a fourth faint Off-mode pixel. Normal setup/deployment installs an unprivileged sampler. Requires updated matrix firmware; active game, T/L, pairing, and startup displays are unchanged.
- Complete hand-setup backups with player name, saved sensitivity, and calibration; restore review offers explicit same-position calibration reuse or fresh centering. Portable backups now start at version 2, as noted above. Restore pauses controller output, preserves Academy progress, and resumes safely after an interrupted calibration write. Existing player stores migrate with retained backups and unchanged progress.
- Idle-only matrix attract setting in Setup: On, Dim, or Off with faint app, console-network, and authenticated RetroPie indicators. Saves without restarting vision; game modes, T, L, startup, errors, and pairing remain unchanged.
- Controller-stored player presets with individual hand sensitivity, saved Academy lessons, and persistent Glove Master awards. Existing settings become Player 1 with a private version-1 backup retained before migration.
- Player switches and restores pause controls; stale tabs cannot overwrite another player or undo progress resets. Device credentials remain excluded from portable backups.
- Exact software commit/candidate metadata and matrix source-fingerprint readback, distinguishing running firmware from the expected packaged sketch. Older firmware is reported as unavailable instead of assumed to match.
- Phone/tablet layout refinements for navigation, forms, player controls, and Academy/Play camera panels, verified at widths down to 320 pixels.

Changed

- Extracted Dashboard, Academy, shared page rendering, Games, and personalization into maintained web modules; removed obsolete tuning UI without changing rendered pages.
- Receiver defaults to signed controller input; an explicit temporary `--allow-legacy-controller` upgrade option closes after the first signed state.
- Reorganized Setup with clearer connection, pairing, attract, and power sections; one-time-code pairing is prominent and SSH password pairing remains available. Address checks explicitly test name resolution; the hand/glove label is identified as diagnostic. Refreshed Help, installation instructions, screenshots, and PDFs.
- Recorded review coverage and a maintainer parking lot for later decisions.

Fixed

- Setup recovers from failed loading and actions, retains unsaved connection fields during controller/pairing refreshes, and resets approval when the pairing host changes.
- Concurrent device settings saves are serialized and privately atomic; first-run configuration is private from creation. Rotating the pairing key disarms output.
- Failed worker delivery of Start/Stop is reported as pending and retried using only the latest request; Dashboard and Setup show failures instead of implying success.
- Supervised worker launches read the pairing token from a private file instead of exposing it in process arguments. Browser mutation routes reject cross-site requests, and connection saves require JSON.
- Receiver socket timeouts release native input as well as the virtual gamepad; malformed nested profile messages no longer terminate the listener.
- Glove Academy now replaces the final lesson with the Glove Master award after all sixteen lessons are completed, instead of displaying both and pushing the lesson downward. Start again restores the lesson panel.

[0.3.2-rc.5] - 2026-09-06

Cumulative release-candidate notes since 0.3.1. This candidate adds the refined idle matrix animation, reproducible live latency baselines, and updated native gameplay validation. Visible camera-to-game latency and a reliable stationary jitter baseline still need a coordinated video test; this is not the final 0.3.2 release. The project base version remains 0.3.2, with the candidate identity recorded by the release tag and installer manifests.

Added

- Added [scripts/measure-vision-status.py](#) for read-only, bounded latency baselines. It ignores duplicate cached status samples, separates changing run conditions, and reports observed timing distributions and optional neutral X/Y variation. Reports contain aggregate measurements, with explicit limits for unmeasured network, receiver, core, and display stages. Recognition and movement defaults are unchanged.
- Recorded separate live movement, requested-stationary, network round-trip, and native-publication cadence observations. The movement window retained detection in all 315 observed samples and measured 132.8 ms read-to-send p95; physical display latency and a reliable stationary-jitter baseline remain separate validation steps.
- Added a camera-controlled Rock Paper Scissors page at [/play](#). A closed hand plays rock, an open palm plays paper, and the held V-sign plays scissors in a first-to-three match against Pixel Pal. It reuses the local practice-camera lease, pauses cabinet input while open, and provides touch and mouse controls. Help now points readers to its illustrated Gameplay Guide section, while This cabinet retains a direct link to the game.
- Confirmed every implemented native Super Glove Ball action during a completed live game: open-hand release/throw, fist grab/catch, index-point Robo-Bullet, and fist-plus-forward Power Punch, alongside Start and continuous X/Y. Movement is playable and substantially improved; residual latency remains the next refinement target. Unused wrist-rotation and packet-button fields stay neutral.

Changed

- Refined the idle matrix animation with a double-flash lightning bolt, separated fingers and thumb, a consistent cuff buckle, a longer fist hold, and a smaller travelling spark. Grayscale shading and a gradual glow finish the four-second loop. Added a preview rendered from the sketch's actual frames; physical LED appearance remains a separate check. Requires a matrix firmware update.
- Updated native compatibility test assertions to match the already documented completed-game confirmation and deliberately neutral unused packet fields.
- Corrected relay-test mock argument access for Python 3.7 compatibility.
- Adopted **PowerGlove Vision Controller** as the user-facing name for the Arduino UNO Q device throughout current guides, while retaining literal `uno-q` commands, filenames, host placeholders, and hardware-specific notes.
- Remembered the player's explicit Start/Stop controller choice across UNO Q application and system restarts. Registered RetroPie launches now maintain a renewable session only while RetroArch is running, allowing an armed controller to reconnect after an application restart without emitting into EmulationStation or the runcommand menu. Stop remains sticky; game exit, unknown games, and stale sessions release controls. Dashboard now distinguishes armed delivery from the live game session.
- Standardized numbered and bulleted lists across the Markdown guides. Built-in Help now keeps wrapped and loosely spaced items together, and generated PDFs keep every marker beside its complete item at page boundaries.
- Connected Super Glove Ball's native open hand, closed hand/fist, index-point, and calibrated signed-Z packet fields. The UNO Q publishes five-finger compound recognition explicitly; the custom Nestopia core emits `$00`, `$FF`, `$0F`, and absolute depth for throw, grab, Robo-Bullet, and fist-plus-forward Power Punch validation. Exact-ROM headless traces verify the ten-byte packet values; wrist rotation and remaining native button codes stay neutral.
- Replaced Glove Academy's threshold-first Tune panel with a Pixel Pal-guided personalization wizard. Families choose the problem, confirm clear framing, record gesture-specific steps, pass a two-use preview test, and then save only affected shared components. Raw numerical controls now live under Advanced.
- Added dedicated **Show your hand** and **Find neutral** artwork, clearer Start and Select lesson titles, and an intentional **Set this as my center** action.
- Added an optional local Academy diagnostic. It exercises the deployed proven tracker, emits an aggregate report without images or landmarks, and deletes its temporary video after analysis, cancellation, or a 30-minute abandonment.
- Added advisory hand-framing, darkness, and backlighting guidance without changing camera exposure automatically.
- Reduced the deliberate V-sign Start hold from 650 ms to 500 ms while retaining its short pulse, interruption cancellation, Menu Guard priority, and 300 ms visible-release rearm guard. Select remains a 150 ms hold.
- Evaluate directions, finger curls, wrist rolls, and Closed Hand on every fresh inference result. Glove Zap and Pull Back now reject stationary near/far hands and one-frame scale jumps by requiring two beyond-threshold observations plus 0.10 normalized palm-scale motion in the intended direction within 250 ms. Existing held, pulsed, turbo, and toggle mappings remain unchanged.
- Added rolling camera-read-to-send and changed-control-to-send p50/p95 readings to Dashboard, including continuous native X/Y changes. The native coordinate benchmark now verifies a deliberate step reaches 90% within 150 ms without adding another queue or core-side smoothing.

- Added local-only fixed and user-paced camera record/replay tools for identical-frame comparisons of MediaPipe Hands thread counts, experimental Tasks Video, 640×480 versus full-field 512×384, and preview-open versus preview-closed cost. The recorder covers near/far, gesture, jitter, depth, and recovery cues. Guided capture keeps a live browser preview visible, records each step only after player confirmation, and releases the camera when complete. Clips are temporary and never used for training.
- On the deployed UNO Q, preliminary 300-sample steady-state windows kept the 640×480 camera-read-to-decision p95 at 127.9 ms with Dashboard closed and 125.7 ms with its stream open. A 16-lane guided replay retained MediaPipe Hands at 640×480 with four threads: Tasks Video roughly doubled inference latency, and 512×384 did not meet the required 15% improvement while producing more false activations. Controller-enabled game measurements remain a separate acceptance step.
- Promoted the Project Overview PDF to the first position in built-in Help's Technical Documentation section and reordered that section from broad system orientation through increasingly specialized implementation material.
- Made the deployed legacy-lite MediaPipe tracker explicit while retaining an opt-in Tasks Video path for controlled comparisons. Camera capture now drains continuously into a newest-frame slot, background JPEG work can drop stale previews, and live diagnostics report frame freshness, inference cadence, skipped captures, and preview cost.
- Named the tracker choices **MediaPipe Hands (proven)** and **MediaPipe Tasks Video (experimental)** in Dashboard and documentation while retaining the stable [legacy](#) and [tasks-video](#) command identifiers.
- Replaced Glove Academy's mislabeled single-finger curl in the **Close your hand** lesson with a dedicated closed-fist illustration. The intentionally six-digit glove remains part of Pixel Pal's Extra-Digit Hunt, and the pose is now shown in the Gameplay Guide, Configuration Reference, quick reference, and Programs A-I manual wherever closed-hand recognition is taught.
- Made **Skip lesson**, **Previous**, and automatic lesson completion mutually consistent. Manual navigation now invalidates older recognition responses and pending automatic advances, so one button press changes the lesson once.
- Added guarded UNO Q recovery for one UVC camera. Installation may run without a camera; the first healthy sighting records the camera and its real parent hub in a root-owned allowlist, and a later healthy sighting updates the association if the camera moves. After a sustained outage the helper validates and resets only the last observed hub. Recovery is rate-limited, never loops during one outage, and leaves ordinary camera reopen attempts as the first response.

[0.3.1] - 2026-09-05

Documentation and Help maintenance release adding Pixel Pal's deliberately spoiler-free guide game and keeping packaged documentation verifiable.

Added

- Added Pixel Pal's Extra-Digit Hunt to the two illustrated gameplay guides. Readers count every preserved six-digit glove once per appearance, then find the verified per-guide answer at the back. Built-in Help keeps the answer behind an explicit reveal, accessible descriptions identify qualifying art, and a checked manifest prevents future artwork changes from making the answer stale. The App Lab package builder now refreshes its tracked checksum companion after each successful build so guide updates cannot leave a stale digest.

Stable feature release covering shared recognition, native Super Glove Ball support, simplified personal tuning, repeatable two-machine installation, and the completed illustrated documentation set.

Native Super Glove Ball and emulator support

- Confirmed the exact Super Glove Ball ROM's ten-byte native packet, MSB-first reads, `$A0/$5F` detection, `$3F` terminator, native `$82` Start, and continuous X/Y behavior in a deterministic headless trace.
- Corrected the custom Nestopia core to wrap the exact ROM's stream at ten bytes, center zero precisely, and neutralize stale, lost, uncalibrated, and wrong-profile samples instead of retaining a prior coordinate.
- Added reversible per-ROM native Nestopia/FCEUmm selection plus a RetroPie-only isolated core installer; stock Nestopia remains untouched.
- Audited all eight listed US ROMs: only Super Glove Ball consumes native multi-byte glove packets; the other games use standard controller-bit mappings through FCEUmm.
- Added a reproducible matched-savestate direction benchmark. Native Super Glove Ball visibly activates and releases every axis by frame 3, including a 3.1% X step; FCEUmm Gun.Smoke polls input on frame 1 and visibly activates and releases every direction by frame 2.
- Compared the exact Super Glove Ball ROM in both cores, confirmed that FCEUmm remains standard-joypad-only, and corrected native Y wrapping so its packet and screen position span bottom, center, and top.
- Added an optional RetroPie installer offer that builds the pinned GPLv2 native core locally, registers both per-ROM launch choices without changing the saved FCEUmm selection, and installs the upstream license beside the core.

Installation and distribution

- Added versioned UNO Q and RetroPie installers with verified downloads, repeatable updates, backups, and read-only checks.
- Included the early-start and shutdown helpers in UNO Q installation; App Lab's command-line tools install the app and Arduino sketch.
- Added optional emulator installation and registered-game checks, including Bad Street Brawler's game-specific Glove Zap setting.
- Simplified installation instructions and moved manual repair and packaging details into the technical reference.
- Added release packaging and automated installer tests after fresh-device and physical gameplay validation on the UNO Q and RetroPie console.
- Added installation manifests shared by both package installers and Wi-Fi deployment. Updates back up and remove unchanged obsolete application files, preserve local edits, and provide interrupted-update recovery and read-only inventory checks.
- Allowed UNO Q deployments to select an explicit SSH identity and expanded the live deployment gate to verify every published technical guide and PDF.
- Made `config/profiles.json` release-owned and replaceable during updates so the shared recognition defaults supersede legacy per-game thresholds; the previous file is backed up and personal calibration and tuning under `data/` remain preserved.
- Hardened release validation so installer packages must contain the shared profile baseline and current recognition, tracking, tuning, and native-core selection modules. Update tests explicitly preserve neutral calibration and personal gesture tuning while replacing release-owned defaults.
- Preserved existing profile configuration during installation and Wi-Fi updates.
- Unified release/deployment payload selection and installer templates; removed unused helpers and state.

Recognition, performance, and safety

- Made explicit and automatic neutral calibration accept only complete hand observations at 70% confidence or better. Documented its 24-frame averaging, circular wrist mean, 95th-percentile jitter measurement, repeatability limits, atomic replacement, and separation from portable recognition defaults.
- Removed avoidable gameplay diagnostics: finger geometry is measured once, detailed landmarks are prepared only at the 5 fps preview cadence, and idle tuning skips measurement work while unchanged configurations are reused.
- Made Menu Guard and the V-sign mutually exclusive with a clear pinky deadband. Menu Guard now has priority, cancels pending Start pulses, and suppresses every ordinary controller button while the safety pose is active.
- Made recognition settings global across game mappings, reduced movement travel with [0.28](#) activation and [0.14](#) release baselines, and recorded neutral X/Y jitter so only noisy setups automatically receive higher safe thresholds.
- Expanded Glove Academy from twelve to sixteen mapping-independent lessons with roll left/right, close hand, and a shared menu-guard pose. Practice polls every 75 ms and controller delivery remains paused.
- Hardened the V-sign Start command after live Gun Smoke/FCEUmm testing exposed accidental pause pulses. Start now requires a stable 0.65-second hold and a continuous 0.30-second non-V release before it can rearm; movement, A/B, and Select timing are unchanged.
- Reduced Start and Select pose debounce to 0.15 seconds before the later live-tested Start-specific hold and rearm protection, retaining one press per pose and release before repeating.
- Fixed controller session resets, malformed packet rejection, and timeout release during rejected network traffic.
- Applied finger release thresholds consistently in Programs E, F, and G.
- Matched tuning feedback to each recording step and made failed matrix commands retry.

Documentation and licensing

- Expanded the third-party runtime inventory for the modified Nestopia core with its patch checksum, affected files, runtime/install boundaries, GPL preservation rules, and pinned-update procedure. Documented stock RetroArch and FCEUmm as externally installed RetroPie dependencies and added an audit check that prevents the Nestopia revision or patch digest from drifting away from the published notices.
- Preserved Nestopia's original source copyright and GPL header verbatim, added a separate file-by-file PowerGlove Vision modification ledger, made the native-core build reject header changes, and installed that ledger beside the optional core and its upstream [COPYING](#) file.
- Centered gesture, profile, and matrix artwork in Help and printable table columns. Grouped related startup and attention-state matrix photographs into compact visual rows where side-by-side comparison is clearer.
- Added a dedicated menu-guard illustration and substantially enlarged the A/D/H experiment artwork in built-in Help and the gameplay PDF, with balanced text columns.
- Added the isolated [lr-nestopia-powerglove](#) research patch, reproducible pinned build, tracing, and guarded latest-sample bridge. Exact-ROM validation is now recorded alongside the retained FCEUmm fallback.

Added

- Added an optional permanent UNO Q early-start helper after a successful physical cold-boot trial. It releases the installed sketch earlier on each boot while preserving the system boot display and existing hourglass animation.
- Added a Matrix display guide in Help and PDF form, covering every display state, profile letters, pairing sequences, troubleshooting, and photographs of the physical display.
- Added a pulsing startup hourglass before Router Bridge initialization. A dedicated Arduino sketch display task keeps it moving while Linux is starting; ordinary app status then selects the glove, profile, or Academy display. The protected system boot display remains unchanged.
- Renamed the Learn section to Glove Academy in navigation, the page heading, and current guides. Kept [/learn](#) links, lesson/tuning behaviour, and L/T matrix indicators unchanged.
- Added startup stage timings for library imports, model preparation, camera initialization, and first inference.
- Added individual gesture illustrations and Pixel Pal to the website and friendly manuals, with a smaller PNG for web use.
- Preserved and bundled the unmodified Google Hand Landmarker model with Apache 2.0 license text, provenance, and checksum.
- Added offline model-cache installation and recovery from the bundled copy, with verified download fallback only when the bundle is absent.
- Made package builds verify the model and required license files before reporting success.

Changed

- Preloaded OpenCV and MediaPipe in the background when the vision worker starts, keeping controls responsive and the camera off until requested. On the tested cabinet, first activation after a reboot took 1.21 seconds once preloading completed.
- Made [/dashboard](#) the default page, retaining redirects from [/](#) and [/debug](#).
- Made web footers use the release version, with a development indicator on dev builds.

Gesture tuning simplification

- Replaced seven recordings with open hand, gesture, open hand (three seconds each).
- Added optional all-finger hand setup using a gentle fist with the thumb outside.
- Added live extended/curled finger feedback sharing V-sign and thumbs-up recognition checks. Existing saved thresholds remain compatible.
- Completed live camera validation on the installed UNO Q system.
- Featured common tuning controls and grouped directions, wrist rolls, and other combinations under More adjustments. Added explicit movement-specific starting-position and return instructions for Glove Zap, pull-back, directions, and wrist rolls.
- Added pull-back as the twelfth ordinary Learn lesson, with a live action indicator, backward-distance feedback, and personal pull thresholds.
- Gameplay movement mappings now use shared activation/release states for wrist controls, forward push, pull-back, movement answers, and braking. Preserved game button mappings, pulses, and pull-toggle edges; clear depth activation on tracking loss.
- Matched Tune's T animation to Learn's L using the same scan line, trailing glow, brightness, and frame timing.
- Standardized Glove Zap and Pull Back labels across Learn, Tune, and the guides.

Documentation and firmware reconciliation

- Documented optional five-finger hand setup, three recordings of three seconds, preview/save/discard/reset, extended-only finger handling, and unchanged version-1 saved settings.
- Updated ordinary Learn to twelve lessons with Glove Zap and Pull Back, and documented shared gameplay activation/release states while retaining profile mappings and menu holds.
- Documented matching scanning L/T animations and consistent action names.
- Recorded the verified Zephyr 1.0.0 platform and pinned sketch libraries, distinguishing platform installation, compile-only validation, and firmware upload.
- Refreshed maintained guides and their PDF editions. Historical entries above retain the behaviour and names at the time of each change.
- Completed live camera, game-control, cold-boot, and physical display checks on the installed systems in addition to the automated release checks.

Complete-pose validation before tuning suggestions

- Automatic analysis now requires all selected fingers to match together in at least 90% of accepted samples, including opening and final release phases.
- V-sign validates extended index/middle alongside curled ring/pinky; thumbs-up validates extended thumb alongside four curled fingers, using shared recognition checks and existing personal extension thresholds.
- Failed analysis identifies the finger/phase and clears the preview without modifying saved settings. Existing holds and orientation-flexible thumb extension remain unchanged.
- Added regressions for invalid extended fingers in each phase, personal thresholds, exact boundaries, noise tolerance, simultaneous pose failures, invalid samples, and saved-value preservation, then completed live camera validation.

Bad Street Brawler Glove Zap

- Mapped forward-push activation to a 180 ms simultaneous Left + Right pulse in the Bad Street Brawler profile. Require release before retriggering; cancel on menu poses, tracking loss, and recalibration.
- Enabled opposing directions in Bad Street Brawler's FCEUmm game options while preserving global settings. The existing RetroPie receiver and gamepad mapping need no changes.
- Corrected the earlier documentation claim that this attack required native glove support. Added pulse, cancellation, rearming, and other-profile regressions; all 155 tests passed. A two-frame headless test confirmed the game-specific options load. Live attack validation remains outstanding.

Repeatable RetroPie Glove Zap configuration

- Added [powerglove-bsb-zap](#) with read-only checks and explicit `--apply` setup.
- Checks installed FCEUmm/RetroArch and game emulator selection; reports missing dependencies or incompatible selection without silently changing either.
- Preserves inherited options and global settings; backs up and atomically updates the game file, with repeat-run detection and rollback instructions.
- Added six isolated tests for creation, preservation, inheritance, missing cores, wrong emulator selection, read-only mode, running games, redirects, and symlinks.
- Documented installation, use, limitations, and the remaining live gameplay check.

[0.2.5] - 2026-09-04

This release adds web game mappings, personal gesture tuning, reliable game-launch profile selection, and refreshed illustrated manuals.

Added

- Added a dedicated T on the UNO Q matrix while gesture tuning is active.
- Added a Games JSON editor with paired RetroPie access, duplicate-name validation, conflict detection, verified saves, backup download, and restoration.
- Added guided Learn tuning with camera measurements, independent gesture thresholds, temporary previews, and persistent personal adjustments shared across profiles.
- Added a confined RetroPie Games service on TCP 55358 and included it in installation and health checks.

Fixed

- Moved game mapping editing into Setup and compacted Learn tuning, placing threshold values beneath the camera.
- Published UDP profile control through a persistent App Lab brick, acknowledged queued requests independently of camera startup, and corrected launch-hook rejection reporting and configuration-error handling.
- Kept profile changes responsive during blocked camera startup or reads, and reused the camera and tracker when switching between active profiles.
- Added exact compressed ROM filenames to the default registry so supported `.zip` and `.7z` games can select their profiles.

Documentation

- Refreshed all six interface screenshots on September 4, 2026, blurring camera imagery before capture.
- Illustrated Setup's Games section and Learn's compact Tune layout across the guides, including the T matrix indicator.
- Regenerated the printable manuals to match the updated Markdown and screenshots.
- Reorganized the overview and installation guide around a complete, numbered setup path: Git download, App Lab package import, UNO Q host setup, RetroPie installation, pairing, and gameplay checks.
- Added Programs A-I controls to the overview and a complete command-line reference with project flags, defaults, arguments, and the system-command options used in the guides.
- Applied reader-focused writing conventions across the guides, added first-round game exercises, and corrected the RetroPie update path. PDF regeneration remains a separate publication step.
- Reviewed all Markdown guides for natural English, replacing sentence fragments and compressed notes with complete explanations while retaining concise tables and release entries.
- Revised the Quick Reference with complete installation prerequisites, camera inspection commands, embedded screenshots, calibration explanations, and step-by-step game registration. Clarified shutdown readiness and local tests.
- Corrected stale lesson counts, profile descriptions, calibration behaviour, pairing guidance, reboot verification status, and receiver removal steps.
- Proofread the Quick Reference and normalized Markdown list indentation and continuation text for consistent rendering.
- Regenerated all ten PDF editions, repaired internal section links, and improved heading and image pagination.

[0.2.0] - 2026-09-03

Added

- Added an app-owned Avahi resolver brick that survives App Lab Compose regeneration, replacing the temporary direct socket mount.
- Added one-command host setup for RetroPie and UNO Q, with managed-file backups, preserved private configuration, delayed startup, and read-only PASS/FAIL/ACTION checks. Empty pairing tokens no longer cause restart loops.
- Added explicit A, B, and GLOVE ZAP practice lessons and live indicators. Learn consistently uses the general profile without changing the selected game.
- Added a Glove Master completion achievement to Learn after all eleven lessons are recognized, with Start again and Dashboard actions. Skips do not count.
- Added hand illustrations to every Learn lesson and finger/pose feedback for Start and Select. Learn accepts confirmed menu poses after their short button pulse ends, rather than requiring a second hold longer than the pulse.
- Added a live active-profile selector to the Dashboard without changing the startup profile saved on Setup.
- Added temporary Learn sessions that start vision while preserving the selected profile and desired controller state, including multi-tab leases and automatic recovery when a page disappears unexpectedly.
- Added a dedicated matrix gestures-idle state with a pinball-style animated glove, separate from both true shutdown and the flashing error X.
- Added a dedicated Learn-mode matrix state with a bright L and moving grayscale scan highlight.
- Added a root-owned tmpfiles rule that restores shutdown-helper readiness after reboot or App Lab application replacement.

Changed

- Shutdown now requests a graceful system halt instead of poweroff, which was observed to reboot the UNO Q. Reinstall the host helper to apply this change; hardware tests subsequently confirmed automatic restart with both the powered hub and direct Mac USB connection. Remaining halted is a known unresolved limitation.
- New installations leave the RetroPie destination blank, show generic hostname examples, and keep local practice available before pairing. Controller start requires a destination; saved destinations survive updates.
- Added persistent host Avahi resolution for `.local` gameplay and pairing destinations, with five-second address refresh and deployment mount setup.
- Standardized Dashboard/Learn Calibrate actions with red busy and blue completed feedback, consistent navigation buttons, and shorter Connection/Shutdown labels.
- Prioritized controller transmission before matrix updates and limited browser JPEG encoding to 15 fps; added `inference_ms` and `send_ms` diagnostics.
- Prepared SSH pairing dependencies separately in the persistent runtime cache so dependency downloads do not consume the SSH connection deadline.
- Tuned thumbs-up/Select closed-finger detection to 0.42 using live pose measurements, retaining the straight-thumb requirement and deliberate hold.
- Tuned V/Start closed ring and pinky detection to 0.42 using live pose measurements, retaining straight index/middle checks and the deliberate hold.

- Tuned curl activation/release to 0.50/0.35 using live comfortable index-curl measurements. Learn now follows the same hysteresis state as gameplay, independent of pulsed buttons. Calibration resets held finger switches.
- Finger curl now uses the strongest joint bend and includes the base knuckle for the four fingers. Learn exposes exact curls, a magnified landmark view, and forward-movement readings. Push lessons use continuous feedback so a one-frame event cannot be missed by browser polling.
- Camera finger curl now uses 3D world landmarks, with an aspect-corrected normalized-depth fallback. Movement and gesture thresholds are unchanged.
- Made Dashboard and Learn show camera/tracker startup with elapsed time and a first-start explanation, without enabling the camera in Gestures off.
- Kept Learn startup feedback updating before camera frames arrive and disabled centring until vision is active.
- Replaced generic Program A-I labels on Dashboard and Setup with the program letter and its intended game or use, while retaining the existing profile IDs.
- Standardized the reader-facing game name "Gun Smoke" throughout Help and the public guides; exact [Gun .Smoke](#) ROM basenames remain unchanged for matching.
- Made leaving Learn restore the selected profile, camera state, and controller state. Loading or refreshing the Dashboard now clears abandoned Learn sessions, prevents their old heartbeats from reactivating vision, and reapplies the selected mode.
- Made Wi-Fi deployments preserve PowerGlove Vision as the UNO Q default startup app so the dashboard returns after a board reboot.
- Made Wi-Fi deployments restore the shutdown readiness marker when the installed host watcher is active.
- Extended deployment health verification to tolerate a three-minute cold App Lab runtime startup.
- Made deployment verification use the UNO Q address from the active SSH connection instead of accidentally selecting a Docker bridge interface.
- Made **Gestures off** a healthy worker state that releases controller input, closes the camera and MediaPipe tracker, and keeps the website and authenticated RetroPie profile listener available.
- Made camera and model initialization lazy so idle mode performs no capture or vision processing and can return to an active profile without restarting the website.
- Reworked the matrix attract sequence into distinct pinball-style beats: a four-frame energy sweep, a broad travelling cuff, a staged glove reveal, intermediate finger curls, an eight-position spark with a comet trail, one outline pulse, and a readable hold.
- Used the UNO Q matrix's full eight-level grayscale range to separate the dim glove body, spark halo, bright spark, and whole-glove pulse.

Documentation

- Documented the UNO Q matrix as an eight-level monochrome DMD/BitPixel-style design target, including silhouette, contrast, motion, pulse, and physical review guidance for future animations.
- Standardized source headers with each file's purpose, author, copyright, SPDX license identifier, local history, and links to the complete history.
- Documented public interfaces and non-obvious security, lifecycle, tracking, packaging, and rendering functions.
- Added this centralized project changelog and its print-ready PDF edition.
- Added an automated audit for required source headers, module descriptions, and production Python interface docstrings.
- Added a complete configuration reference covering JSON templates and active copies, gesture thresholds, manifests, RetroArch mapping, systemd units, generated state, secret handling, and the App Lab installation ZIP location.
- Added GitHub Actions checks for supported Python versions, tests, source and documentation audits, syntax, PDF builds, and App Lab installation ZIP verification.
- Added a security policy for private reporting, pairing and network boundaries, shared-token handling, shutdown permissions, and dependency integrity.
- Added a contributing guide for code style, tests, documentation, changelog, generated artifacts, commits, and pull requests.
- Added reusable documentation and App Lab installation package verification tools.
- Added an illustrated, one-page-per-game handbook for all eight automatically configured titles, with original direction, finger-pose, wrist, and depth art.
- Placed cropped gesture illustrations beside the corresponding instructions in the gameplay handbook, including the universal V-sign and thumbs-up controls.
- Added the same contextual gesture illustrations to every Program A-I card and documented the menu gestures shared by all nine programs.
- Added an off-script gameplay section that encourages safe experiments with other NES and Famicom games, especially the unassigned A, D, and H programs.
- Updated the cabinet cheat sheet with off-script profile testing, safe behaviour for unknown games, and exact ROM registration guidance.
- Renamed the built-in installation Help route, retained the original URL as an alias, and aligned its summaries with the Play Checklist terminology.
- Extended the documentation audit to require Help-library coverage and a gameplay section for every title in the shipped game registry.
- Expanded Wi-Fi deployment verification to confirm the current installation and gameplay guides, raw Markdown, and gesture artwork on the UNO Q.
- Fixed the Help renderer so the allowlisted gesture images embedded in control tables appear alongside their actions, while unsafe raw HTML remains escaped.
- Extended UNO Q deployment checks across every Help guide and representative gameplay and Programs A-I illustrations.

- Documented the required Markdown-to-PDF workflow and UNO Q Help synchronization steps for documentation contributions.
- Added offline PDF links to every public Help guide and the project overview, while keeping the cabinet-specific quick-reference PDF private.
- Included the nine allowlisted public PDFs in UNO Q deployments and App Lab installation packages, with package and live-route verification.
- Established `dev` as the integration branch, documented release and hotfix promotion into `main`, and enabled CI validation for pushes to both branches.
- Refreshed the Dashboard, Learn, and Setup screenshots from the running UNO Q after the tagline, compact diagnostics, and safe-shutdown controls were added.
- Added an offline Help library that renders the maintained public Markdown guides with responsive navigation, contents links, illustrations, tables, code samples, and access to the original source.
- Added a dynamic **This cabinet** Help page whose UNO Q links follow the browser's validated hostname or IP and whose non-secret RetroPie values come from the active configuration.
- Rewrote the configuration reference for public installations, with guided UNO Q and RetroPie setup, field-level behaviour, safe gesture tuning, network boundaries, backup and recovery advice, and symptom-based troubleshooting.
- Renamed the Field Guide as the Installation Guide and generalized camera instructions for UVC-compatible USB cameras while recording the Razer Kiyo as tested reference hardware.

Fixed

- Changed Wi-Fi deployment to use SFTP staging and terminal-backed remote commands for UNO Q systems that stall non-terminal SSH sessions.
- Allowed the UNO Q deployment health check to use the board's current IP when its `.local` name pauses during a container restart.
- Published the host shutdown request atomically so a filesystem observer cannot consume the request between file creation and the final content write.
- Updated the GitHub Actions workflow to use the current Node 24 action releases and run the Python 3.7 compatibility job on Ubuntu 22.04.
- Corrected Program I so index curl accelerates in Knight Rider, a forward push accelerates with turbo, and thumb curl fires the weapons.

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Added

- Camera-only Power Glove tracking on Arduino UNO Q with MediaPipe.
- Gesture profiles for Bad Street Brawler, Super Glove Ball, and cartridge-free Programs A-I.
- Authenticated UDP profile selection and virtual Linux gamepad output for RetroPie, including per-game runcommand hooks.
- Dashboard, live diagnostics, offline gesture lessons, configuration controls, camera recovery, controller start/stop controls, and UNO Q matrix feedback.
- Wi-Fi deployment, App Lab packaging, runtime-asset retrieval, branded PDF generation, and a fixed-purpose host shutdown helper.
- Project overview, installation guide, quick reference, profile handbook, third-party component notice, screenshots, and reproducible build instructions.

Changed

- Delayed RetroPie receiver startup until EmulationStation finishes its initial device scan, preventing the virtual controller from disrupting cabinet input.
- Made the controller start disarmed and kept tracking/dashboard operation alive when RetroPie or the camera temporarily becomes unavailable.
- Moved private device settings and downloaded model data outside the App Lab installation ZIP.
- Changed the App Lab installation ZIP to download Google's Hand Landmarker model on first launch and verify its pinned SHA-256 digest before installation.

Fixed

- Recovered cleanly from USB camera disconnects and slow UVC camera wake-up.
- Kept the vision loop responsive through receiver, DNS, Wi-Fi, and mDNS loss.
- Corrected password pairing so credentials never appear in command arguments and the shared token is transferred and installed reliably.
- Corrected UNO Q dependency isolation, secure token upload, PDF builder file mode, landscape diagnostics layout, and cabinet launch integration.

Security

- Added short-lived TLS pairing with certificate comparison, a physical single-use PIN, bounded handshakes, and restricted token-file permissions.
- Required confirmation and a fixed host-side request path for system shutdown.
- Added a third-party component notice covering licenses, provenance, pinned versions, checksums, and update procedure.

Neutral calibration retention

The app now saves completed calibration so the reference survives transitions between Learn and gameplay, profile changes, camera reconnections, and restarts. When you recalibrate, the app replaces the saved reference atomically.

RetroPie mDNS installation

The RetroPie setup command now installs `avahi-daemon` and `libnss-mdns`, enables Avahi at boot, and checks both the service and dependency. Existing hostname configuration and pairing settings are preserved. Fresh-machine installation has not yet been tested.

The UNO Q installer also installs and checks `libnss-mdns` alongside Avahi for host-level resolution, while retaining the separate app-container resolver.

Documentation and shutdown wording consistency

The documentation now explains when to recalibrate, which files to back up, and what the hand-identification score means. Shutdown confirmations also explain that the UNO Q may restart automatically.